Bahria University, Islamabad

Department of Software Engineering

Artificial Intelligence Lab (Spring-2026)

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Lab Journal: 03

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Task No:	Task Wise Marks		Documentation Marks		Total Marks (20)
	Assigned	Obtained	Assigned	Obtained	
1	3		5		
2	3				
3	3				
4	3				
5	3				

Comments:

Signature

Lab No: 03 – OOP in Python

Introduction

Introduction to Object-Oriented Programming in Python.

Tools Used

- Python
- VS Code
- Github

Link to Code: https://github.com/TALHA-089/AI-LAB-03.git

Task 1:

CODE:

Task1.py > ...

```python
from abc import ABC, abstractmethod
import math


class Shape(ABC):
    @abstractmethod
    def calculate_area(self):
        pass


class Rectangle(Shape):
    def __init__(self, length, width):
        self.length = length
        self.width = width

    def calculate_area(self):
        return self.length * self.width


class Square(Shape):
    def __init__(self, side):
        self.side = side

    def calculate_area(self):
        return self.side ** 2


class Circle(Shape):
    def __init__(self, radius):
        self.radius = radius

    def calculate_area(self):
        return math.pi * (self.radius ** 2)


class Cylinder(Shape):
    def __init__(self, radius, height):
        self.radius = radius
        self.height = height

    def calculate_area(self):
        return 2 * math.pi * self.radius * (self.radius + self.height)


if __name__ == "__main__":
    rect = Rectangle(10, 5)
    square = Square(4)
    circle = Circle(7)
    cylinder = Cylinder(3, 8)

    print(f"Rectangle Area: {rect.calculate_area()}")
    print(f"Square Area: {square.calculate_area()}")
    print(f"Circle Area: {circle.calculate_area():.2f}")
    print(f"Cylinder Area: {cylinder.calculate_area():.2f}")

```

Figure 1. Code for Task 1

Screenshot (Output):

```
Talhas-MacBook-Air:Lab 03 talha$ python3 Task1.py
Rectangle Area: 50
Square Area: 16
Circle Area: 153.94
Cylinder Area: 207.35
Talhas-MacBook-Air:Lab 03 talha$
```

Figure 2. Output for Task 1

Task 2:

Code:

```
Task2.py > ...
def calculate_square(x): return x ** 2


if __name__ == "__main__":
    number = 6
    print(f"The square of {number} is {calculate_square(number)}")

```

Figure 3. Code for Task 2

Screenshot (Output):

```
Talhas-MacBook-Air:Lab 03 talha$ python3 Task2.py
The square of 6 is 36
Talhas-MacBook-Air:Lab 03 talha$
```

Figure 4. Output for Task 2

Task 3:

Code:

```
Task3.py > ...
You, 12 minutes ago | 1 author (You)
def get_string_length(s): return len(s)


if __name__ == "__main__":
    sample_string = "Artificial Intelligence"
    print(
        f"The length of the string '{sample_string}' is {get_string_length(sample_string)}")

```

Figure 5. Code for Task 3

Screenshot (Output):

```
Talhas-MacBook-Air:Lab 03 talha$ python3 Task3.py
The length of the string 'Artificial Intelligence' is 23
Talhas-MacBook-Air:Lab 03 talha$
```

Figure 6. Output for Task 3

Task 4:

Code:

Task4.py > ...

```python
import random


def random_number_generator():
    for _ in range(5):
        yield random.randint(1, 100)


if __name__ == "__main__":
    print("5 Random Numbers:")
    for num in random_number_generator():
        print(num)

```

Figure 7. Code for Task 4

Screenshot (Output):

```
Talhas-MacBook-Air:Lab 03 talha$ python3 Task4.py
5 Random Numbers:
12
25
98
31
5
Talhas-MacBook-Air:Lab 03 talha$
```

Figure 8, Output for Task 4

Task 5:

Code:

```
Task5.py > ...
def fibonacci_generator(n):
    a, b = 0, 1
    while a <= n:
        yield a
        a, b = b, a + b


if __name__ == "__main__":
    limit = 50
    print(f"Fibonacci sequence up to {limit}:")

    for num in fibonacci_generator(limit):
        print(num)

```

Figure 9. Code for Task 5

Screenshot (Output):

```
Talhas-MacBook-Air:Lab 03 talha$ python3 Task5.py
Fibonacci sequence up to 50:
0
1
1
2
3
5
8
13
21
34
Talhas-MacBook-Air:Lab 03 talha$
```

Figure 10. Output for Task 5

Conclusion:

In conclusion, this lab provided practical experience with advanced object-oriented programming concepts in Python. Through the exercises, we successfully implemented class inheritance and abstract methods to build a shape hierarchy.